

Trainings ▾

General Information

All Delco Remy America alternators are rated for inlet air temperatures of -35 to 93°C [-31 to 200°F], measured from 25.4 mm [1 in] from the rear of the unit, except for the 26SI. The 26SI has a maximum rating of 85°C [185°F] because of its sealed electronics. Avoid mounting the alternator near any heat sources, such as manifolds, turbochargers, and so forth. Adequate clearance **must** also be provided at the rear of the alternator for cooling air to enter (the alternator fan pulls air through toward the front of the unit).

The second most common cause of failure of Cummins alternators is contamination by dirt, chemicals, water, or unknown substances. While the alternator is laboratory tested to withstand salt spray, dust, and high humidity, it is necessary to mount the alternator to minimize environmental and application contamination. Exposure to salt spray over an extended period of time will shorten the alternator life, as well as increase circuit resistance by causing corrosion or terminal connections for all electrical components. Exposure to engine oil or oil mist from the engine breather tubes can cause brush problems in brush alternators, as well as attract other debris. Any contamination of the rear of the alternator can reduce airflow, which eventually overheat the diodes.

If operating in extreme environments, especially off-road, the 21SI can be installed with a dust shield, which has been shown to reduce the effects of extreme environments and improve alternator life. Debris shields have been developed for the 19SI, 22SI, and 23SI alternators. New debris shields have provisions for mounting an air duct in order to route clean air where necessary.

Establish proper cable routing, and determine cable lengths. Avoid routing taut cables between connections. The maximum voltage drop in the charging system at rated alternator output is 0.5 VDC. For a 12 VDC system, 1.0 VDC for a 24 VDC system, and 1.5 VDC for a 32 VDC system. The voltage drop in the charging system **must** be kept to a minimum in order to make sure enough voltage exists to charge the batteries.

While all Delco Remy America alternators are negative grounds, it is suggested that a full copper return be provided to the negative (-) terminal of the batteries. This will also make sure that as the vehicle ages, corrosion of the frame and ground strap do **not** cause excessive voltage drops. Also, painted mounting brackets can prevent a complete electrical circuit. Be sure all electrical connections, including ground connections, are metal-to-metal connections.

⚠ CAUTION ⚠

Never use any type of toothed lock washer, internal or external tooth (also called star washers), on wiring connections. The points of the star washer cause current to be conducted only through the points of the star washer, rather than the larger flat surface of a standard flat washer, and lead to corrosion and a high-resistance electrical connection.

- All capscrews **must** be a minimum of 12.7 mm [0.5 in].
- **Always** use a hardened steel washer under mounting capscrew heads, or use flanged-head hardware. The washer **must** match the capscrew's diameter.
- **Never** use any type of lock washer, including toothed or split lock washers, on any mounting connection. Lock washers are **not** recommended for mounting capscrews because their trapezoidal cross section provides less clamping surface and higher clamping stress.

The operating range of the cranking motor and solenoid is -25 to 121°C [-13 to 250°F]. Military application starters have an operating range from -54 to 121°C [-65 to 250°F]. Locating the motor close to the exhaust manifold or other heat source can result in reduced motor performance with deterioration of insulation and rubber components. A minimum 50.8 mm [2 in] clearance is recommended. Where relocation or rotation is **not** possible, metal shielding can effectively be used to protect the starter from engagement in hot conditions.

Establish proper routing. Avoid routing a taut cable between connections. When routing battery cables from the frame rails to the cranking motor, anchor the cables to the engine, transmission, or cranking motor, and allow for relative movement between the engine and the frame. It is highly recommended that a copper ground return be used and the frame **not** be relied upon for ground return in the starting circuit, which will prevent corrosion from creating excessive voltage drops. Using a copper ground cable is especially critical in applications using a wet flywheel housing, where reliance on frame grounding can create a current path through the crankshaft and main bearings. Limit the number of cables attached to the battery terminal of the solenoid to **not** more than two.

When positioning the motor at the mounting location, it is desirable to place the solenoid above the centerline of the motor (preferably a minimum of 15 degrees above the horizontal). This will prevent potential accumulation of foreign material or moisture that can restrict operation in the tower portion of the shift lever housing. It also reduces the possibility of road debris being thrown up and breaking the solenoid cap.

- The flange-to-flywheel distance is the distance on the engine from the cranking motor mounting surface to the ring gear face. This distance 50.8 ± 1.52 mm [2.0 ± 0.060 in]. This dimension **must** include spacers or gaskets used on the engine. The cranking motor is designed to function within this dimension. If it is altered, pinion or ring gear damage can occur.
- Delco Remy America recommends the use of 12-point, flanged head capscrews.
- No lock washers, star washers, or soft washers, including copper electrical connections, can be used. Lock washers and star washers do **not** have a rectangular cross section and can result in a loss of clamp force on the mounting capscrews. The yield strength of soft washers or copper connections can be exceeded with mounting capscrews tightened. In each case, a loose starter mounting will result.

⚠ CAUTION ⚠

If the engine does not start and must be cranked again, always wait two to three seconds to allow the pinion and ring gear to stop. If the starter is reengaged immediately, the pinion can be milled or the armature twisted. The starter is intended only for cranking the engine and must not be used to “bump” the engine.

If the engine does **not** start within five seconds of cranking, follow the cold-start procedure in the corresponding Cummins operation and maintenance manual. Operating the cranking motor for more than 30 seconds at a time causes undue stress and/or severe internal heat buildup. A minimum 2-minute cooling period is necessary between extended cranking cycles.

When mounting a magnetic switch on a vehicle, the axis of the plunger **must** be horizontal to the ground and perpendicular to the axis of travel of the vehicle. This is to make sure that the switch is **not** activated by the movement of the vehicle. Do **not** mount the switch on the engine or any metal that can possibly resonate as a result of road or engine vibration. For engines with separate fuel shutoff solenoids, Cummins requires the use of two magnetic switches: one for the fuel shutoff circuit and one for the starting circuit. The fuel shutoff solenoid **must not** be wired to the “S” terminal of the starting motor. Wiring to the “S” terminal will cause the shutoff solenoid to fail.